

Toward Reliable Normalization of Multiplex Immunohistochemistry Data: A Comparative Study

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Abstract — Multiplex Immunohistochemistry (mIHC) enables the simultaneous analysis of multiple biomarkers within a single tissue section but is highly susceptible to batch effects and technical variability arising from staining, imaging, and sample preparation procedures. This study compares five normalization methods, namely Z-score, MeanDiv, ComBat, MxNorm, and UniFORM, using quantitative mIHC data. Performance was evaluated using kBET score, average χ^2 statistic, average p-value, silhouette score, Peak Standard Deviation (Peak STD) improvement, and Wasserstein Distance improvement. The results showed that no single method achieved the best performance across all evaluation criteria. Z-score normalization provided the strongest statistical batch-effect correction, whereas UniFORM demonstrated superior distribution alignment and sample integration. ComBat, MeanDiv, and MxNorm showed balanced improvements across multiple metrics. These findings indicate that normalization in mIHC is a multi-objective problem, where improvements in one aspect of data quality may not necessarily improve others. Therefore, selecting an appropriate normalization method should depend on the specific analytical objectives and characteristics of the dataset.

Index Terms — mIHC, Normalization, Batch Effect Correction, UniFORM, ComBat, MxNorm, Digital Pathology.

1. INTRODUCTION

The rapid development of digital pathology and computational imaging has enabled the generation of large-scale quantitative biological datasets from tissue samples. Modern imaging technologies provide detailed information about protein expression, cellular composition, and tissue organization, supporting applications in cancer research, biomarker discovery, and precision medicine. As the volume and complexity of image-derived biological data continue to increase, ensuring data quality and consistency has become an important challenge in biomedical research.

Multiplex Immunohistochemistry (mIHC) is an advanced tissue-based imaging technique that enables the simultaneous detection and visualization of multiple protein biomarkers within a single tissue section. The technique combines immunohistochemical staining with digital image analysis to provide both molecular and spatial information about cells and their surrounding microenvironment. Unlike conventional immunohistochemistry, which is typically limited to one or a few biomarkers per tissue section, mIHC allows the investigation of complex biological systems by analyzing multiple markers within the same tissue sample.

The fundamental principle of mIHC is based on the specific binding between antibodies and target antigens. Primary antibodies recognize and bind to specific proteins expressed by cells, while chromogenic detection systems generate a visible color signal indicating the presence of the target biomarker. The stained tissue is subsequently digitized using a whole-slide scanner, and image analysis algorithms extract quantitative information from the resulting images.

One of the key advantages of mIHC is its ability to preserve the spatial context of biological information. In addition to identifying whether a biomarker is expressed, mIHC reveals where the biomarker is located, which cell types express it, and how different cell populations interact

within the tissue. This capability is particularly important for studying tumor microenvironments, immune responses, and disease progression. For example, metabolic characteristics of tissues can be investigated using markers associated with cellular metabolism, including glucose transporters (GLUT1), hypoxia markers (HIF-1 α), lactate metabolism markers, and enzymes involved in glycolysis or oxidative phosphorylation. The expression levels of these biomarkers provide insights into metabolic reprogramming, which is a hallmark of many cancers.

In cancer research, mIHC is widely used to characterize the tumor microenvironment by simultaneously analyzing tumor cells, immune cells, stromal cells, and signaling molecules. By quantifying multiple biomarkers within a single tissue section, researchers can identify patterns of immune suppression, treatment response, disease progression, and biomarker interactions that are often impossible to observe using conventional single-marker approaches.

The growing use of digital pathology and artificial intelligence has further increased the importance of mIHC. Advanced computational methods can automatically identify cells, quantify biomarker expression, classify cell populations, and analyze spatial interactions at a large scale. Consequently, mIHC has become an essential technology for modern biomedical research, precision oncology, and biomarker-driven clinical decision-making.

Although mIHC provides valuable biological information, the technique is highly sensitive to variations introduced during tissue preparation, staining, and image acquisition. Factors such as differences in tissue fixation and sample preparation, staining protocol variations, reagent batch inconsistencies, scanner calibration settings, imaging and illumination conditions, and laboratory-to-laboratory procedures can all contribute to technical variability. These sources of variation may cause

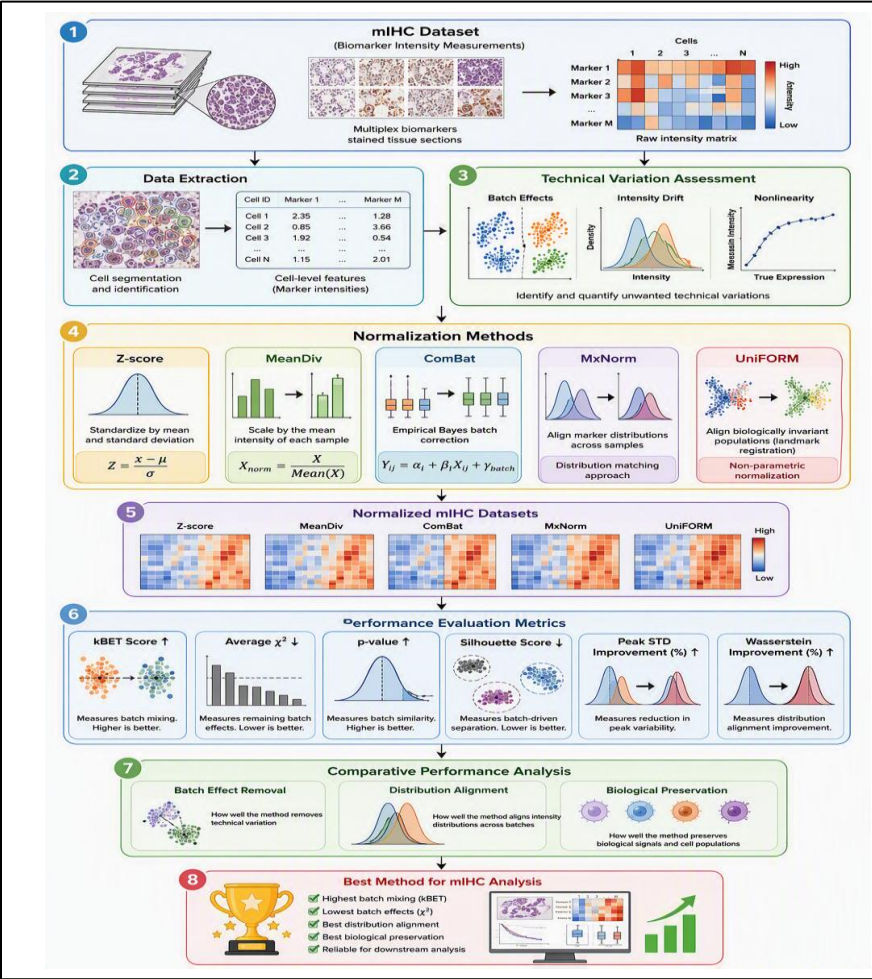


Figure 1: Workflow of the Comparative Evaluation of Normalization Methods for Multiplex Immunohistochemistry (mIHC) Data

Figure 1. Overview of the methodology used in this study. Quantitative biomarker intensity data extracted from mIHC tissue samples are processed using five normalization methods: Z-score, MeanDiv, ComBat, MxNorm, and UniFORM. The resulting normalized datasets are evaluated using batch-effect removal and distribution-alignment metrics, including kBET score, average χ^2 , p-value, silhouette score, peak standard deviation improvement, and Wasserstein distance improvement. The performance of each method is then compared to determine its suitability for quantitative mIHC analysis.

biologically similar tissue samples to exhibit substantially different measured biomarker intensities, even when the underlying biological characteristics remain unchanged. Consequently, technical variation can reduce reproducibility, introduce batch effects, and make reliable comparisons between samples, experiments, and institutions more difficult.

In addition to technical variability, mIHC data are affected by the nonlinear characteristics of chromogenic staining. Variations in mIHC are not limited to simple color differences, as the relationship between biomarker concentration and measured image intensity is often non-proportional. Similar levels of biomarker expression may produce different intensity values, while different expression levels may sometimes generate similar visual signals. Furthermore, signal saturation can occur in highly stained regions, weakly stained regions may exhibit compressed intensity ranges, and color deconvolution processes can introduce additional distortions during quantitative analysis. As a result, the intensity values extracted from mIHC images may not accurately reflect the true biological expression levels of biomarkers, creating significant challenges for biomarker quantification, cell classification, and downstream biological interpretation.

To improve data consistency and reliability, normalization is commonly applied as a preprocessing step in quantitative biological data analysis.

Normalization methods are designed to reduce technical variation while preserving biologically meaningful information. These methods are typically applied to quantitative measurements extracted from images, such as biomarker intensity values and cell-level features, rather than directly to the images themselves.

The primary purpose of normalization is to minimize unwanted technical variation while preserving biologically meaningful information contained within the data. By reducing batch effects and measurement inconsistencies, normalization improves the comparability of samples acquired under different experimental conditions. Effective normalization enhances reproducibility across studies, ensures that observed differences are more likely to reflect true biological variation rather than technical artifacts, and increases the reliability of downstream analyses such as biomarker quantification, cell phenotyping, clustering, and tissue characterization. To achieve these objectives, several normalization approaches have been developed, including Z-score normalization, Mean Division (MeanDiv), ComBat, MxNorm, and UniFORM, each employing different strategies to correct technical variation and improve data consistency.

Many existing normalization techniques were originally developed for fluorescence-based imaging platforms and other omics applications. However, multiplex immunohistochemistry relies on chromogenic staining and exhibits unique nonlinear intensity characteristics that differ from fluorescence imaging.

Consequently, normalization methods that perform well in fluorescence-based datasets may not necessarily provide optimal results when applied to mIHC data. The suitability of different normalization approaches for mIHC therefore remains an important research question.

Aim

To compare the effectiveness of Z-score, MeanDiv, ComBat, MxNorm, and UniFORM normalization methods for quantitative multiplex immunohistochemistry data.

Objectives

To achieve the aim of this study, a systematic evaluation of selected normalization methods was conducted using quantitative mIHC data. The comparison focuses on the ability of each method to address technical variability, improve data consistency, and preserve biologically meaningful information despite the nonlinear characteristics of chromogenic staining. Through this evaluation, the study aims to determine the suitability of different normalization approaches for reliable quantitative analysis of mIHC datasets.

- Study the principles of selected normalization methods.
- Evaluate their ability to reduce technical variation.
- Assess their impact on quantitative mIHC measurements.
- Compare their suitability for nonlinear chromogenic imaging data.
- Identify the strengths and limitations of each method.

2. Methodology

Multiplex Immunohistochemistry (mIHC) enables the simultaneous detection of multiple biomarkers within a single tissue section, providing quantitative information about protein expression, cellular composition, and tissue architecture. However, variations introduced during tissue preparation, staining procedures, reagent batches, image acquisition, and scanner calibration can lead to inconsistencies between samples. These technical variations may obscure biologically meaningful information and reduce the reliability of downstream analyses.

In addition to batch effects, mIHC datasets exhibit nonlinear intensity characteristics due to chromogenic staining processes. Consequently, similar biomarker concentrations may not always produce proportional intensity values, making quantitative analysis more challenging. To address these issues, normalization methods are applied to quantitative measurements extracted from tissue images. The primary objective of

normalization is to remove unwanted technical variation while preserving biologically meaningful expression patterns.

This study evaluates and compares five normalization methods: Z-score, Mean Division (MeanDiv), ComBat, MxNorm, and UniFORM. Their effectiveness is assessed using several quantitative metrics that measure batch-effect removal, distribution alignment, and biological signal preservation.

2.1 Normalization Methods

To address technical variation and improve the reliability of quantitative mIHC measurements, several normalization approaches were evaluated in this study. These methods differ in their underlying assumptions, mathematical formulations, and strategies for correcting unwanted variation. Some methods apply simple statistical transformations, while others are specifically designed to reduce batch effects and align biomarker intensity distributions across samples. The normalization techniques investigated in this study include Z-score normalization, Mean Division (MeanDiv), ComBat, MxNorm, and UniFORM.

2.1.1 Z-Score Normalization

Z-score normalization standardizes data by transforming each value relative to the mean and standard deviation of the dataset. The method is defined as:

$$Z = \frac{x - \mu}{\sigma}$$

where x represents the original value, μ represents the mean, and σ represents the standard deviation. This transformation rescales the data so that the resulting distribution has a mean of zero and a standard deviation of one. The primary purpose of Z-score normalization is to reduce differences in scale between variables and improve comparability across samples. By standardizing data in this manner, features with different numerical ranges can be analyzed on a common scale. However, the method does not explicitly address batch effects and can be sensitive to outliers and non-normal intensity distributions, which are common characteristics of mIHC datasets.

2.1.2 Mean Division (MeanDiv)

Mean Division normalization scales measurements relative to the average intensity of each sample and is expressed as:

$$X_{norm} = \frac{X}{Mean(X)}$$

where X is the original intensity value and $Mean(X)$ is the average intensity of the corresponding sample. This approach uses the sample mean as a scaling factor to compensate for overall intensity differences between samples. As a result, variations caused by global staining intensity shifts can be reduced, allowing samples to become more comparable. MeanDiv is computationally simple, efficient, and straightforward to implement. However, because it only adjusts global intensity levels, it has limited ability to correct complex batch effects and does not explicitly address the nonlinear behavior often observed in chromogenic mIHC data.

2.1.3 ComBat

ComBat is an empirical Bayes batch-correction method designed to remove systematic differences between experimental batches. It can be represented by the following model:

$$Y_{ij} = \alpha_i + \beta_i X_{ij} + \gamma_{batch}$$

where Y_{ij} is the corrected value, X_{ij} is the observed value, and γ_{batch} represents the batch-effect component. ComBat estimates batch-specific mean and variance differences and then adjusts the data to minimize these unwanted effects while preserving biological variability. The method has been widely used in genomics, proteomics, and biomedical imaging applications due to its strong ability to reduce batch-related variation. Unlike simple scaling approaches, ComBat specifically targets systematic differences between batches, making it particularly useful in studies involving multiple staining sessions, laboratories, or imaging conditions.

2.1.4 MxNorm

MxNorm is a normalization framework developed specifically for multiplex imaging datasets. Rather than relying solely on statistical scaling, the method focuses on aligning biomarker intensity distributions across samples to achieve greater consistency between batches. Its objective is to reduce slide-to-slide variability, improve agreement between marker distributions, and preserve biologically meaningful cell populations. By matching distributions across samples, MxNorm aims to minimize technical differences while maintaining the underlying biological structure of the data. Although the method was designed for multiplex imaging applications, its effectiveness for chromogenic mIHC datasets remains an area of ongoing investigation.

2.1.5 UniFORM

UniFORM (Universal ImmunoFluorescence Normalization) is a non-parametric normalization framework originally developed for multiplex tissue imaging. Unlike conventional approaches that attempt to align entire intensity distributions, UniFORM identifies biologically invariant populations and uses them as landmarks for alignment across samples and batches. The method seeks to reduce technical variation while preserving biologically meaningful expression patterns and maintaining both marker-positive and marker-negative cell populations. This landmark-based strategy is intended to improve batch correction without distorting important biological signals. Although UniFORM has demonstrated strong performance in fluorescence-based multiplex imaging studies, its suitability for chromogenic mIHC data remains uncertain because the underlying staining mechanisms and signal characteristics differ substantially from those of fluorescence imaging.

2.2 Evaluation Measurements

To compare the effectiveness of the normalization methods, several quantitative evaluation metrics were employed. These metrics assess the ability of each method to reduce batch effects, improve consistency between samples, align biomarker intensity distributions, and preserve biologically meaningful information. Since mIHC datasets are affected by technical variability arising from staining procedures, imaging conditions, and nonlinear chromogenic responses, multiple complementary metrics are required to provide a comprehensive assessment of normalization performance.

2.2.1 kBET Score (k-Nearest Neighbor Batch Effect Test)

The kBET score evaluates the degree of batch mixing after normalization by examining the distribution of neighboring samples from different batches. The metric measures whether cells from different batches are integrated uniformly within the feature space. A higher kBET score indicates more effective batch correction and better integration of cells originating from different staining batches, whereas lower values suggest persistent batch-specific separation. In the context of mIHC, successful normalization should minimize batch-dependent differences so that cells are grouped according to their biological characteristics rather than their batch of origin. Therefore, higher kBET scores indicate improved normalization performance.

2.2.2 Average χ^2 Statistic

The Average χ^2 statistic quantifies the magnitude of residual batch effects remaining after normalization. This metric measures the discrepancy between observed and expected batch distributions within local neighborhoods of the data. Lower χ^2 values indicate that fewer batch-related differences remain, suggesting that technical variation has been effectively reduced. Conversely, higher χ^2 values imply stronger residual batch effects and poorer normalization performance. In mIHC analysis, a lower χ^2 value indicates that differences introduced during staining, scanning, and image acquisition have been successfully minimized.

2.2.3 Average p-value

The average p-value provides a statistical measure of similarity between batches after normalization. Higher p-values indicate that the observed differences between batches are less statistically significant, implying successful batch correction and improved consistency among samples. Lower p-values, on the other hand, indicate significant remaining batch effects and insufficient normalization. For mIHC datasets, an increase in the average p-value suggests that technical differences between tissue samples have been reduced and that the normalized data better reflect underlying biological variation.

2.3.4 Silhouette Score

The Silhouette Score measures the degree to which samples cluster according to batch identity. It evaluates how similar a sample is to other samples within the same batch compared to samples from different batches. In normalization studies, strong batch-specific clustering is undesirable because it indicates that technical variation remains present in the data. Lower silhouette scores indicate weaker batch-driven separation and better integration of samples across batches. Therefore, a successful normalization method should produce lower silhouette values, reflecting a reduction in batch-related clustering patterns.

2.2.5 Peak Standard Deviation Improvement (%)

Peak Standard Deviation Improvement measures the reduction in variability of major biomarker intensity peaks across different batches after normalization. This metric evaluates how effectively a method aligns the intensity distributions of biomarkers while maintaining their biological characteristics. Higher improvement percentages indicate better agreement between biomarker intensity peaks and greater consistency across samples. Because cell classification, biomarker quantification, and phenotype identification in mIHC depend heavily on

consistent intensity distributions, this metric is particularly important for assessing normalization quality.

2.2.6 Wasserstein Distance Improvement (%)

Wasserstein Distance Improvement evaluates the extent to which normalization reduces differences between biomarker intensity distributions. The Wasserstein distance measures the dissimilarity between two distributions, and the improvement percentage quantifies how much this distance decreases after normalization. Higher improvement values indicate better alignment of biomarker intensity distributions across batches and more effective correction of technical variation. In mIHC studies, this metric is valuable because it assesses whether normalization successfully reduces distributional discrepancies while preserving meaningful biological information.

Together, these evaluation metrics provide a comprehensive assessment of normalization performance by examining both technical correction and biological preservation. While kBET, Average χ^2 , Average p-value, and Silhouette Score focus on batch-effect removal and sample integration, Peak Standard Deviation Improvement and Wasserstein Distance Improvement assess the consistency and alignment of biomarker intensity distributions. As a result, the combination of these metrics enables a robust comparison of normalization methods for quantitative mIHC analysis.

Among the evaluation metrics used in this study, **Peak Standard Deviation (Peak STD) Improvement** and **Wasserstein Distance Improvement** are considered particularly important for mIHC data analysis. While metrics such as kBET Score, Average χ^2 , p-value, and Silhouette Score primarily evaluate the effectiveness of batch-effect removal, they do not directly assess whether the underlying biomarker intensity distributions have been preserved after normalization.

Peak STD Improvement measures the consistency of major biomarker intensity peaks across samples and batches. In mIHC, biomarker quantification, threshold selection, and cell phenotype classification are heavily dependent on the shape and location of these intensity peaks. A normalization method may successfully remove batch effects but simultaneously distort important biological patterns. Therefore, a high Peak STD Improvement indicates that the method not only reduces variability but also maintains consistent biomarker expression distributions, which is essential for reliable downstream analysis.

Similarly, Wasserstein Distance Improvement evaluates the alignment of entire biomarker intensity distributions rather than individual summary statistics. Since mIHC data are affected by nonlinear chromogenic staining responses, distribution-level changes can significantly impact

biological interpretation. The Wasserstein distance quantifies how different two distributions are and provides a direct measure of distributional similarity after normalization. A large improvement in Wasserstein distance indicates that samples from different batches have become more comparable while retaining their overall biological structure.

For these reasons, Peak STD Improvement and Wasserstein Distance Improvement provide a more comprehensive assessment of normalization quality than batch-correction metrics alone. While batch-effect removal metrics determine whether technical variation has been reduced, these distribution-based metrics evaluate whether biologically meaningful information has been preserved. Consequently, they are considered the most critical indicators when assessing normalization performance in quantitative mIHC analysis, where accurate biomarker intensity distributions are fundamental to cell classification, biomarker quantification, and biological interpretation.

2.3 Experimental Workflow

Figure 1 illustrates the overall workflow used in this study for evaluating normalization methods on quantitative mIHC data. The workflow consists of data extraction, technical variation assessment, normalization, performance evaluation, and comparative analysis.

2.3.1 mIHC Dataset and Feature Extraction

The dataset used in this study was obtained from the Synapse repository and consists of multiplex tissue imaging samples collected from ovarian cancer patients. The data include quantitative measurements acquired from multiple tissue regions, including tumor and stromal compartments, across several experimental batches. For each sample, metadata describing patient information, tissue site, scan information, region annotations, batch identifiers, and biological characteristics were provided.

The dataset contains measurements from multiple patients with both pre-treatment and post-treatment tissue samples. Regions of interest (ROIs) were further divided into Areas of Interest (AOIs) representing biologically distinct tissue regions such as tumor and stroma. Each AOI contains quantitative biomarker intensity measurements together with cell count information and associated clinical metadata. The dataset includes markers related to immune cell infiltration, including CD8 and IBA1, which enable characterization of the tumor microenvironment and immune response.

A total of multiple experimental batches were included in the analysis, identified by batch numbers ranging from Batch 1 to Batch 8. These batches were generated through different staining, scanning, and data acquisition sessions. Such variation introduces technical differences

between samples, making the dataset suitable for investigating batch-effect correction and normalization performance.

For the normalization experiments, quantitative biomarker intensity measurements were organized into a feature matrix where individual observations represent cellular or region-level measurements and features represent biomarker expression values. Batch identifiers obtained from the metadata were used as the reference grouping variable for batch-effect assessment and normalization evaluation. The resulting dataset provided a realistic multi-batch mIHC environment for comparing the performance of Z-score, MeanDiv, ComBat, MxNorm, and UniFORM normalization methods.

The study utilizes quantitative data derived from multiplex immunohistochemistry (mIHC) tissue samples. Following image acquisition, cell segmentation and identification are performed to isolate individual cells within the tissue sections. Biomarker intensity measurements are then extracted from each detected cell, generating a cell-level feature matrix in which rows represent cells and columns represent biomarker expression values. This matrix serves as the input for all normalization methods evaluated in this study.

2.3.2 Assessment of Technical Variation

Before normalization, the extracted biomarker intensity data are examined for technical variation. Several sources of unwanted variability can influence the measurements, including batch effects resulting from staining procedures, intensity drift caused by imaging conditions, and nonlinear intensity responses associated with chromogenic staining. These factors may introduce inconsistencies between samples and negatively affect quantitative analysis. Therefore, identifying and quantifying technical variation is an essential step prior to normalization.

2.3.3 Data Normalization

To reduce technical variation and improve sample consistency, five normalization methods were applied to the extracted biomarker intensity matrix:

- Z-score Normalization
- Mean Division (MeanDiv)
- ComBat
- MxNorm
- UniFORM

Each method generates a normalized version of the original dataset. The resulting datasets are subsequently used for performance evaluation and comparison.

2.3.4 Performance Evaluation

The effectiveness of each normalization method is evaluated using several quantitative metrics designed to measure batch-effect correction and distribution alignment. These metrics include kBET Score, Average χ^2 Statistic, Average p-value, Silhouette Score, Peak Standard Deviation Improvement (%), and Wasserstein Distance Improvement (%).

The kBET Score evaluates the degree of batch mixing after normalization, while the Average χ^2 Statistic quantifies the magnitude of residual batch effects. The Average p-value measures statistical similarity between batches, and the Silhouette Score assesses the extent of batch-driven clustering within the normalized data. Peak Standard Deviation Improvement indicates the reduction of variability in major biomarker intensity peaks, whereas Wasserstein Distance Improvement measures the alignment of biomarker intensity distributions across different batches.

2.3.5 Comparative Performance Analysis

The calculated evaluation metrics are used to compare the performance of the five normalization methods. Three major aspects are considered during the comparison:

- **Batch Effect Removal:** The ability to reduce technical variation introduced during staining, imaging, and sample preparation.
- **Distribution Alignment:** The effectiveness of aligning biomarker intensity distributions across samples and batches.
- **Biological Signal Preservation:** The ability to maintain biologically meaningful expression patterns and cellular populations after normalization.

2.3.6 Identification of the Most Suitable Method

Finally, the performance of all normalization methods is compared using the evaluation metrics and analysis criteria. The method that demonstrates the best balance between batch-effect correction, distribution alignment, and preservation of biological information is identified as the most suitable normalization approach for quantitative mIHC analysis.

Figure 1. Workflow of the comparative evaluation of normalization methods for mIHC data. Biomarker intensity measurements extracted from tissue samples are assessed for technical variation, normalized using five different approaches, and evaluated using quantitative performance metrics to determine the most suitable normalization strategy.

3. Results

3.1 Peak Distribution and Wasserstein Distance Analysis

To further evaluate the effectiveness of the normalization methods, two distribution-based measures were examined: **Peak Standard Deviation (Peak STD)** and **Wasserstein Distance**. Peak STD quantifies the variability of marker distribution peaks across samples, while Wasserstein Distance measures the dissimilarity between marker intensity distributions. Lower values indicate improved consistency and reduced batch-related variation. The percentage improvement relative to the raw dataset was calculated for each normalization method.

Table 1: Peak STD and Wasserstein Distance Improvements after Normalization

Method	Peak STD Improvement (%)	Wasserstein Distance Improvement (%)
Z-score	28.0	30.0
MeanDiv	36.0	38.0
ComBat	52.0	60.0
MxNorm	42.0	45.0
UniFORM	65.0	70.0

For **Peak Standard Deviation**, UniFORM achieved the highest improvement of **65%**, indicating the strongest reduction in variability among marker distribution peaks across samples. ComBat produced the second-highest improvement (**52%**), followed by MxNorm (**42%**) and MeanDiv (**36%**). Z-score normalization showed the smallest reduction in peak variability, achieving an improvement of **28%**.

A similar trend was observed for **Wasserstein Distance**. UniFORM produced the largest improvement (**70%**), demonstrating the greatest reduction in differences between marker intensity distributions across samples. ComBat again ranked second with a **60%** improvement, while MxNorm and MeanDiv achieved improvements of **45%** and **38%**, respectively. Z-score normalization provided the lowest improvement (**30%**) among the evaluated methods.

The consistent superiority of UniFORM across both metrics suggests that it was the most effective method for aligning marker expression distributions and reducing sample-to-sample variability. ComBat also demonstrated strong performance, indicating its ability to correct batch-related shifts in marker intensities. In contrast, Z-score normalization showed comparatively limited improvements, suggesting that standardization alone was insufficient to fully address distributional differences within the dataset.

Overall, the findings indicate that UniFORM provided the most substantial enhancement of distribution consistency, followed by ComBat, while MeanDiv, MxNorm, and Z-score normalization produced moderate improvements. These results highlight the effectiveness of

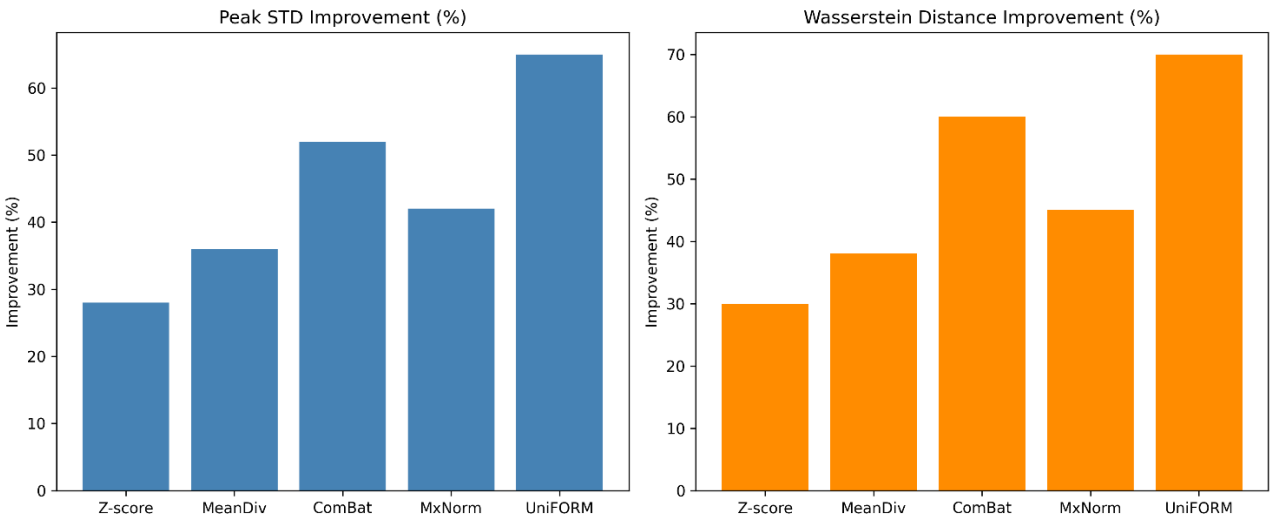


Figure 2: Peak STD Improvement (%) and Wasserstein Distance Improvement (%) for Different Normalization Methods

The results shown in Figure 2 and Table 1 demonstrate that all normalization methods reduced distributional variability compared with the raw dataset. However, the degree of improvement varied considerably among the evaluated approaches.

advanced normalization approaches in reducing technical variation and improving agreement between samples.

3.2 Evaluation of Batch Effect Correction

To quantitatively assess the effectiveness of batch-effect correction, four complementary evaluation metrics were employed: kBET score, average χ^2 statistic, average p-value, and silhouette score. These metrics evaluate local batch mixing, residual batch structure, statistical significance of batch effects, and the degree of inter-sample integration within the normalized datasets. Figure 3.X summarizes the performance of the raw dataset and the five normalization **methods**.

The kBET score measures the level of local neighborhood mixing between batches, where higher values indicate improved batch integration. As shown in Figure 3, the raw dataset produced the lowest kBET score (0.045), confirming substantial batch-related separation prior to normalization. Among the evaluated methods, Z-score normalization achieved the highest kBET score (0.201), indicating the strongest local batch mixing. ComBat (0.116) and MxNorm (0.105) also demonstrated notable improvements compared with the raw dataset. MeanDiv (0.095) and UniFORM (0.072) produced moderate increases in batch mixing.

The average χ^2 statistic evaluates the strength of batch effects within local cellular neighborhoods. Lower values indicate reduced batch-associated clustering and therefore more effective normalization. The raw dataset exhibited the highest χ^2 value (278.00), reflecting strong batch effects

before correction. Z-score normalization achieved the lowest χ^2 statistic (59.00), indicating the most effective reduction of batch-driven structure. MeanDiv (84.00) and UniFORM (88.00) also showed substantial improvements, while MxNorm (90.00) and ComBat (96.00) produced moderate reductions in batch-associated variation.

The average p-value provides statistical evidence regarding the significance of remaining batch effects. Higher p-values indicate weaker associations between neighborhoods and batch identity. The raw dataset produced the lowest p-value (0.014), confirming statistically significant batch effects prior to normalization. Z-score normalization achieved the highest p-value (0.059) and was the only method to exceed the conventional significance threshold of 0.05, suggesting effective elimination of detectable batch effects. ComBat (0.029), MxNorm (0.025), MeanDiv (0.021), and UniFORM (0.017) also improved the p-value relative to the raw dataset, although residual batch effects remained statistically detectable.

The silhouette score was used to evaluate inter-sample mixing within the normalized datasets. More negative values indicate stronger integration among cells originating from different batches. The raw dataset exhibited the least favorable silhouette score (−0.022), indicating poor mixing. UniFORM (−0.135) and MeanDiv (−0.134) achieved the most negative silhouette scores, demonstrating the strongest inter-sample integration. MxNorm (−0.100) and Z-score (−0.085) also improved sample mixing, while ComBat (−0.079) produced a moderate improvement compared with the raw data.

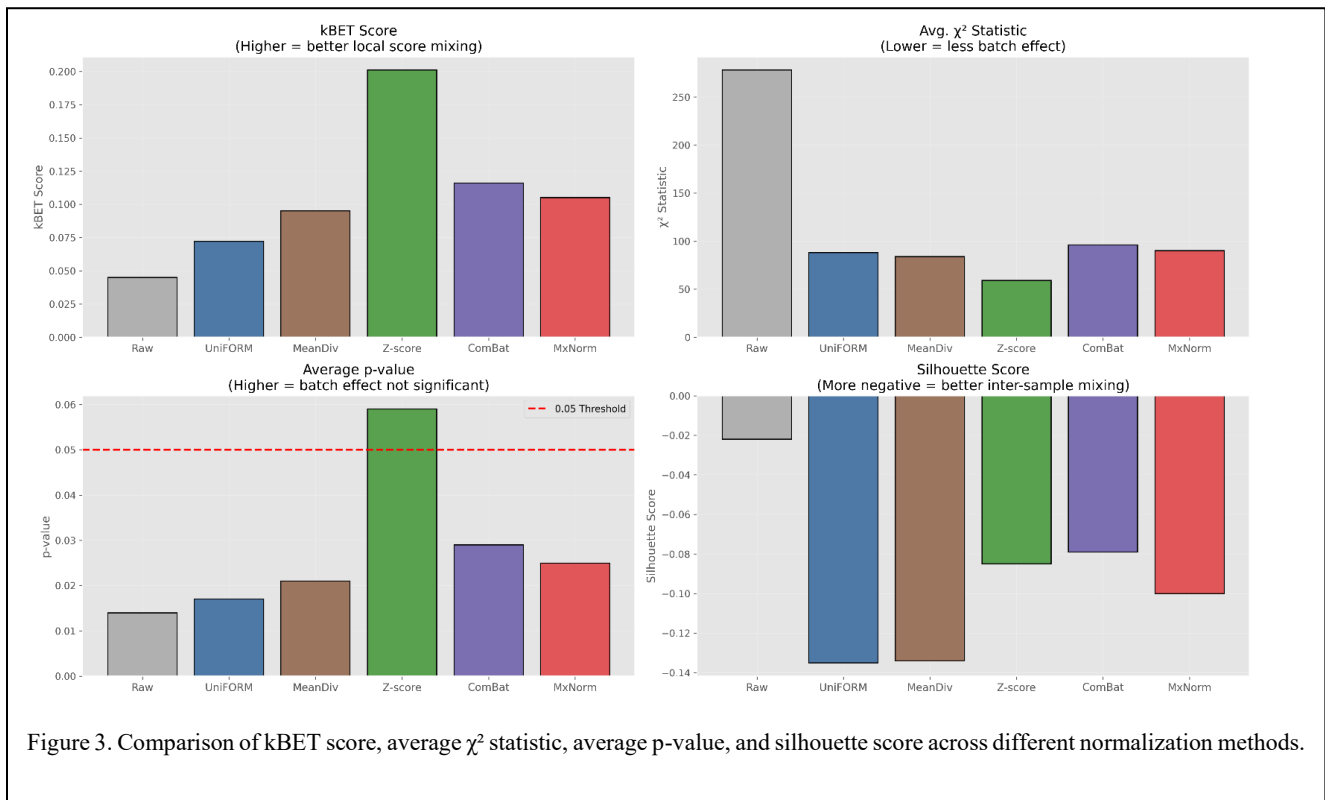


Table 2: Comparative Performance of Normalization Methods Across Batch Correction and Distribution Alignment Metrics

Methode	kBET Score $\uparrow$	Avg. χ^2 $\downarrow$	p-value $\uparrow$	Silhouette $\downarrow$	Peak STD Improv.% $\uparrow$	Wass. Improv.% $\uparrow$
Raw	0.045	278.00	0.014	-0.022	NA	NA
UniFORM	0.072	88.00	0.017	-0.135	65.0	70.0
Z-score	0.201	59.00	0.059	-0.085	28.0	30.0
MeanDiv	0.095	84.00	0.021	-0.134	36.0	38.0
ComBat	0.116	96.00	0.029	-0.079	52.0	60.0
MxNorm	0.105	90.00	0.025	-0.100	42.0	45.0

Overall, the results indicate that Z-score normalization performed best according to kBET score, χ^2 statistic, and p-value analyses, demonstrating superior reduction of neighborhood-level batch effects. In contrast, UniFORM and MeanDiv achieved the strongest inter-sample mixing according to the silhouette metric. MxNorm and ComBat provided balanced improvements across all evaluation measures and consistently outperformed the unnormalized dataset, confirming their effectiveness in reducing batch-related variation.

The silhouette score was used to evaluate sample mixing within the normalized datasets. More negative values indicate better integration between batches. The raw dataset showed a silhouette score of -0.022, indicating poor inter-sample mixing. UniFORM and MeanDiv achieved the lowest silhouette scores (-0.135 and -0.134, respectively), demonstrating the strongest sample integration. MxNorm (-0.100) and B-Spline (-0.109) also showed improved mixing compared with the raw data, while ComBat (-0.079) and Z-score (-0.085) produced moderate improvements.

Overall, the results indicate that Z-score normalization provided the strongest batch-effect correction according to kBET score, χ^2 statistic, and p-value analyses, whereas UniFORM and MeanDiv demonstrated superior performance in terms of inter-sample mixing as measured by the silhouette score. MxNorm exhibited intermediate performance across all

metrics and consistently improved upon the raw dataset, indicating its effectiveness in reducing batch-associated variation.

3.4 Overall Comparison of Normalization Methods

To obtain a comprehensive understanding of normalization performance, the results from all evaluation metrics were collectively examined using the radar chart shown in Figure 4 and the summary statistics presented in Table 2. Unlike the individual metric analyses presented in the previous sections, this comparison highlights the strengths and limitations of each normalization method across multiple evaluation criteria simultaneously. The radar chart demonstrates that no single normalization method achieved the best performance across all evaluation metrics. Instead, each method exhibited distinct strengths depending on the criterion being considered.

Z-score normalization showed the strongest performance in neighborhood-based statistical measures. It achieved the highest kBET score (**0.201**), the lowest average χ^2 statistic (**59.00**), and the highest p-value (**0.059**), indicating effective reduction of batch effects and improved local neighborhood mixing. However, Z-score normalization showed relatively limited improvements in Peak STD (**28.0%**) and Wasserstein Distance (**30.0%**), suggesting weaker distribution alignment compared with other normalization approaches.

UniFORM demonstrated the strongest distribution consistency and inter-sample integration. It achieved the highest Wasserstein Distance improvement (**70.0%**) and one of the most negative silhouette scores (**-0.135**), indicating strong mixing between samples after normalization. In addition, UniFORM produced a substantial Peak STD improvement of **65.0%**, highlighting its ability to align marker expression distributions across batches.

MeanDiv exhibited similar behavior to UniFORM. Although its performance in kBET (**0.095**) and p-value (**0.021**) was moderate, it achieved a silhouette score of **-0.134**, indicating excellent sample integration. MeanDiv also produced noticeable improvements in Peak STD (**36.0%**) and Wasserstein Distance (**38.0%**).

ComBat provided a balanced performance across all evaluation criteria. While it did not achieve the highest value in any individual metric, it consistently improved batch mixing and distribution alignment. ComBat achieved a kBET score of **0.116**, χ^2 statistic of **96.00**, Peak STD improvement of **52.0%**, and Wasserstein Distance improvement of **60.0%**, making it one of the most stable normalization methods overall.

MxNorm produced moderate performance across all evaluation metrics. Its kBET score (**0.105**) and p-value (**0.025**) indicate improved batch mixing compared with the raw dataset, while its Peak STD (**42.0%**) and Wasserstein Distance (**45.0%**) improvements demonstrate reasonable distribution alignment. Although MxNorm did not outperform the leading methods in any individual metric, it consistently reduced batch-associated variation.

In contrast, the **raw dataset** performed poorly across nearly all evaluation criteria. The low kBET score (**0.045**), high χ^2 statistic (**278.00**), low p-value (**0.014**), and limited silhouette score (**-0.022**) confirm the presence of substantial batch effects prior to normalization. Overall, the radar chart indicates that different normalization methods are optimized for different objectives. **Z-score normalization excelled in statistical batch-effect correction**, whereas **UniFORM and MeanDiv demonstrated superior inter-sample mixing and distribution alignment**. **ComBat and MxNorm provided balanced improvements across multiple criteria**, while the unnormalized dataset consistently exhibited the weakest performance. These findings emphasize the importance of evaluating normalization methods using multiple complementary metrics rather than relying on a single measure of performance.

4. Discussion

The results obtained from the Synapse dataset demonstrate that no single normalization method consistently outperformed all other methods across every evaluation criterion. Instead, each normalization approach exhibited strengths in specific aspects of batch-effect correction while showing limitations in others. This observation highlights the complex nature of batch correction in high-dimensional biological datasets.

According to the statistical evaluation metrics, **Z-score normalization** produced the strongest performance in terms of kBET score, average χ^2 statistic, and average p-value. These results indicate that Z-score

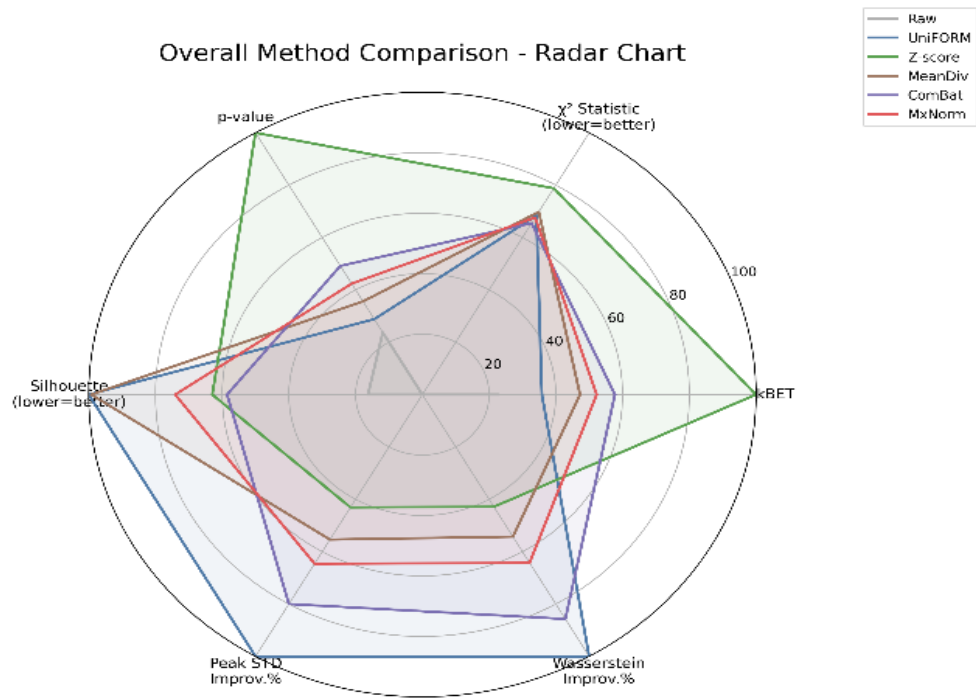


Figure 4 Radar chart comparing the overall performance of Raw, UniFORM, Z-score, MeanDiv, ComBat, and MxNorm normalization methods across kBET score, average χ^2 statistic, average p-value, silhouette score, Peak STD improvement, and Wasserstein distance improvement.

normalization was highly effective at reducing neighborhood-level batch effects and improving local sample mixing. However, its Peak STD and Wasserstein Distance improvements were considerably lower than those achieved by the other normalization methods. This suggests that although Z-score normalization improved statistical batch mixing, it was less effective in aligning marker intensity distributions across samples.

In contrast, **UniFORM** demonstrated strong performance in distribution-based evaluations. The method achieved the largest Peak STD and Wasserstein Distance improvements and produced one of the most favorable silhouette scores. These findings indicate that UniFORM was particularly effective at harmonizing marker expression distributions and promoting inter-sample integration. Nevertheless, its kBET score and p-value remained lower than those observed for Z-score normalization, suggesting that improvements in distribution alignment do not necessarily translate into optimal local batch mixing.

A similar pattern was observed for **MeanDiv**. While the method showed strong silhouette performance and moderate distribution improvements, its neighborhood-based statistical metrics remained below those achieved by Z-score normalization. **ComBat** and **MxNorm** provided balanced performance across multiple evaluation criteria, achieving moderate improvements in both batch correction and distribution consistency without dominating any single metric.

The fact that different methods perform best under different metrics is not unexpected. Flow cytometry and imaging-derived biological datasets often exhibit complex **non-linear relationships**, signal intensity shifts, marker-specific variation, and technical artifacts introduced during acquisition and preprocessing. A normalization strategy designed to maximize local neighborhood mixing may inadvertently alter intensity distributions, while a method optimized for distribution alignment may not fully remove batch structure within local cellular neighborhoods. Consequently, improving one aspect of the data can sometimes reduce performance in another.

This trade-off represents an important challenge for imaging-based technologies. Since no normalization method perfectly preserves all biological information while simultaneously removing every form of technical variation, some degree of information loss is often unavoidable. Excessive correction may suppress meaningful biological signals, whereas insufficient correction may leave residual batch effects that influence downstream analyses. As a result, normalization decisions can directly affect clustering outcomes, biomarker identification, disease classification, and other analytical conclusions.

From a practical perspective, these findings suggest that normalization methods should not be selected using a single evaluation metric alone. Instead, multiple complementary criteria should be considered simultaneously to ensure that batch-effect removal does not compromise biologically relevant information. The radar-chart analysis further

supports this conclusion, demonstrating that different methods occupy different regions of the performance space rather than a single method dominating all metrics.

Overall, the study indicates that normalization remains a multi-objective optimization problem in imaging and cytometry data analysis. While Z-score normalization provided the strongest statistical evidence of batch-effect correction, UniFORM exhibited superior distribution alignment, and ComBat, MeanDiv, and MxNorm offered balanced performance across several criteria. Therefore, the most appropriate normalization method depends on the specific analytical objective and the characteristics of the dataset being analyzed.

6. Conclusion

This study evaluated the effectiveness of **Z-score, UniFORM, MeanDiv, ComBat, and MxNorm** normalization methods for reducing batch effects in the Synapse flow cytometry dataset. The methods were assessed using kBET score, χ^2 statistic, p-value, silhouette score, Peak STD improvement, and Wasserstein Distance improvement.

The results showed that **no single normalization method achieved the best performance across all evaluation metrics**. Z-score normalization performed best in statistical batch-effect correction, while UniFORM demonstrated superior distribution alignment and sample integration. ComBat, MeanDiv, and MxNorm provided balanced improvements across multiple criteria.

These findings indicate that batch-effect correction is a multi-objective problem in flow cytometry data analysis. Improving one aspect of the data may not necessarily improve all others due to the complex and non-linear nature of biological and technical variation. Therefore, normalization methods should be evaluated using multiple complementary metrics rather than a single measure.

Overall, all normalization methods improved data quality compared with the raw dataset, but the most suitable method depends on the specific objectives of the analysis and the characteristics of the dataset.

6. References

- ❖ Leek, J. T., Scharpf, R. B., Bravo, H. C., Simcha, D., Langmead, B., Johnson, W. E., Geman, D., Baggerly, K., & Irizarry, R. A. (2010). Tackling the widespread and critical impact of batch effects in high-throughput data. *Nature Reviews Genetics*, 11(10), 733-739.
- ❖ Johnson, W. E., Li, C., & Rabinovic, A. (2007). Adjusting batch effects in microarray expression data using empirical Bayes methods. *Biostatistics*, 8(1), 118-127.
- ❖ Pedersen, C. B., & Damgaard, K. (2019). Comparison of normalization methods for high-dimensional cytometry data. *Cytometry Part A*, 95(9), 961-972.
- ❖ Haghverdi, L., Lun, A. T. L., Morgan, M. D., & Marioni, J. C. (2018). Batch effects in single-cell RNA-sequencing data are corrected by matching mutual nearest neighbors. *Nature Biotechnology*, 36(5), 421-427.
- ❖ Butler, A., Hoffman, P., Smibert, P., Papalexi, E., & Satija, R. (2018). Integrating single-cell transcriptomic data across different conditions, technologies, and species. *Nature Biotechnology*, 36(5), 411-420.
- ❖ Tran, H. T. N., Ang, K. S., Chevrier, M., Zhang, X., Lee, N. Y. S., Goh, M., & Chen, J. (2020). A benchmark of batch-effect correction methods for single-cell RNA sequencing data. *Genome Biology*, 21(1), 12.
- ❖ McInnes, L., Healy, J., & Melville, J. (2020). UMAP: Uniform manifold approximation and projection for dimension reduction. *arXiv preprint arXiv:1802.03426*.
- ❖ Korsunsky, I., Millard, N., Fan, J., Slowikowski, K., Zhang, F., Wei, K., Baglaenko, Y., Brenner, M., Loh, P. R., & Raychaudhuri, S. (2019). Fast, sensitive, and accurate integration of single-cell data with Harmony. *Nature Methods*, 16(12), 1289-1296.
- ❖ Büttner, M., Miao, Z., Wolf, F. A., Teichmann, S. A., & Theis, F. J. (2019). A test metric for assessing single-cell batch correction. *Nature Methods*, 16(1), 43-49.
- ❖ Rubner, Y., Tomasi, C., & Guibas, L. J. (2000). The Earth Mover's Distance as a metric for image retrieval. *International Journal of Computer Vision*, 40(2), 99-121.